MAHATMA GANDHI UNIVERSITY

MG-UGP (HONOURS)REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

DSC-MG4DSCECT203-WIRELESS TECHNOLOGY

Part A, Answer all questions, 1 mark each

1. (c)PM
2. (b)Wireless communication
3. (c)3Khz-300GHz
4. (c) Multipath propagation and mobility
5. (c)Shadow fading
6. (c) Signal to noise ratio
7. (b) Balanced Mixer (Double Balanced Mixer is the correct answer)
8. (a) Modulator , Amplifier and Antenna
9. (b) IEEE 802.11
10. (c) Signal to noise ratio SNR

Part B , Answer any 4 questions 5 marks each

11. (Definition 2 Mark, Proper explanation of given points 2 Mark, Waveform 1 mark)

A carrier signal in communication electronics is a high-frequency signal used to carry information (message signal) from a transmitter to a receiver.

Definition

A carrier signal is typically a sinusoidal waveform (like a sine wave) that by itself contains no information, but it becomes useful when we modify it using a message signal.

Why we need a carrier signal

Low-frequency signals (like audio) cannot travel long distances efficiently. So we:

- Combine the message signal with a high-frequency carrier
- This process is called modulation

Basic form of a carrier signal

(Waveform)

A carrier signal can be represented as:

$c(t)=A\sin(2\pi ft+\phi)$

Where:

- A = Amplitude
- f = Frequency (high frequency)
- ϕ= Phase

12. (Definition 2 Mark, Proper explanation of given points 3 Marks)

The RF (Radio Frequency) spectrum is the range of electromagnetic frequencies used for wireless communication.

Definition

The RF spectrum refers to the portion of the electromagnetic spectrum with frequencies ranging from about:

3 kHz to 300 GHz

These frequencies are used to transmit signals through the air without wires.

13. (Equation 2 mark, Calculation 2 mark, Correct answer with unit 1 Mark)

Signal-to-Interference Ratio (SIR) is given by:

SIR=Signal Power / Interference Power
SIR=10 mW/2 mW=5

In decibels (dB)

SIR in dB=$10\log10(5)\approx6.99$ dB≈7 dB

14. (Definition 2 Mark, Proper explanation of given points 3 Marks

A link budget is a calculation used in communication systems to determine the received signal strength at the receiver by accounting for all gains and losses in a transmission path.

Definition

A link budget is the balance sheet of power in a communication link, showing how much signal power is transmitted, how much is lost, and how much finally reaches the receiver.

Basic idea

Received Power=Transmitted Power+Gains−Losses

Components of a link budget

Transmitter Power (Pt) – power sent from transmitter

Antenna Gains (Gt, Gr) – gain of transmitting and receiving antennas

Path Loss – loss due to distance (free space loss)

Other Losses – atmospheric loss, cable loss, fading, etc.

Received Power (Pr) – final power at receiver

Expanded formula (in dB)

Pr=Pt+Gt+Gr−Lp−Lother

15. (List of pattern components 1 mark, Explanation of all components 3 marks, Radiation pattern 1 mark)

The radiation pattern of an antenna describes how an antenna radiates or receives electromagnetic energy in different directions in space.

Definition

A radiation pattern is a graphical representation of the variation of radiated power with direction at a fixed distance from the antenna.

Types of Radiation Patterns

1. 3D Radiation Pattern

- Shows radiation in all directions (three-dimensional view)
- Looks like a sphere, doughnut, or complex shape depending on antenna type
- Example: A simple dipole has a doughnut-shaped pattern

2. 2D Radiation Pattern

- A cross-sectional view of the 3D pattern
- Usually plotted in:
    - Polar form (most common)
    - Rectangular form

- Easier to analyze and widely used

Important Terms

Main Lobe

- The region of maximum radiation
- Direction where antenna radiates strongest

Side Lobes

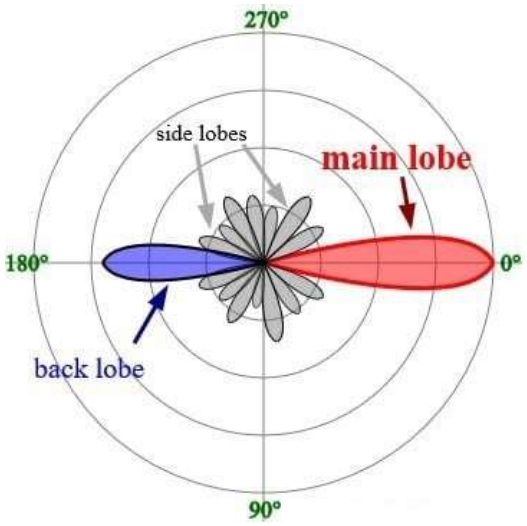
- Smaller lobes in other directions
- Represent undesired radiation

Back Lobe

- Radiation in the opposite direction of main lobe

Nulls

- Directions where radiation is zero or minimum



Beamwidth

Half Power Beamwidth (HPBW)

- Angle between points where power drops to half (−3 dB) of maximum
- Indicates directivity of antenna

Narrow beamwidth = highly directional antenna
Wide beamwidth = less directional

16. (Explanation of all filters 5 mark, any one 1 ¼ Mark)

Basic types of RF (Radio Frequency) filters are classified based on how they allow or block frequency components:

1. Low Pass Filter (LPF)

- Allows low frequencies to pass
- Attenuates high frequencies
- Used for noise reduction and smoothing

2. High Pass Filter (HPF)

- Allows high frequencies to pass
- Blocks low frequencies
- Used to remove DC or low-frequency noise

3. Band Pass Filter (BPF)

- Allows a specific range (band) of frequencies
- Rejects frequencies below and above the band
- Widely used in communication receivers

4. Band Stop Filter (BSF) / Notch Filter

- Rejects a specific band of frequencies
- Allows frequencies outside that band
- Used to eliminate interference (e.g., unwanted signals)

Part C, Answer any 2 questions, 10 marks each

17.(Correct explanation of both spectrum 2 mark, anyone- 1 mark, examples for both 1 mark, Advantages and disadvantages 2 marks)

Licensed Spectrum Bands

These are frequency bands that require official permission (license) from regulatory authorities (like TRAI in India).

Features

- Allocated to specific users (telecom operators, broadcasters)
- Exclusive use → minimal interference
- Requires payment (auction/fees)
- Highly regulated

Examples of Use

- Mobile networks (2G, 3G, 4G, 5G)
- TV broadcasting
- Satellite communication

Advantages

- Reliable communication
- Better quality (less noise/interference)
- Secure and controlled

Disadvantages

- Expensive
- Limited availability

Unlicensed Spectrum Bands

These bands can be used without a license, but must follow certain rules.

Features

- Open for public use
- Shared spectrum → more interference possible
- No licensing cost
- Limited power restrictions

Examples of Use

- Wi-Fi (2.4 GHz, 5 GHz)
- Bluetooth
- Cordless phones

Advantages

- Free to use
- Easy access
- Supports innovation (IoT devices, etc.)

Disadvantages

- Interference and congestion
- Less reliable
- Limited range/power

18. (Correct Explanation of given points 5 marks)

The bandwidth–data rate trade-off is a key concept in modern wireless communication, especially when using higher-order modulation schemes.

Basic Idea

To transmit data faster (higher data rate), you generally need:

- More bandwidth, OR
- More bits per symbol (using advanced modulation)

But increasing one often affects the other.

Role of Modulation

Lower-order modulation (e.g., BPSK, QPSK)

- Fewer bits per symbol
- Requires more bandwidth for high data rates
- Very robust (works well in noisy conditions)

Higher-order modulation (e.g., 16-QAM, 64-QAM, 256-QAM)

- More bits per symbol
- Achieves higher data rate within the same bandwidth
- Less robust (more sensitive to noise and interference)

Key Relationship

Data Rate=Bandwidth ×Spectral Efficiency

Spectral efficiency (bits/s/Hz) increases with higher-order modulation

- So, higher-order modulation → more data in same bandwidth

Trade-Off Explained

1. Bandwidth Efficiency vs Noise Immunity

- Higher-order modulation:
  - High data rate
  - Requires high Signal-to-Noise Ratio (SNR)
- Lower-order modulation:
  - Reliable communication
  - Lower data rate

2. Power Requirement

- Higher-order schemes need more transmit power to maintain accuracy
- Otherwise, error rate increases

3. Error Performance

- As modulation order increases:
  - Symbol points get closer
  - System becomes more prone to errors

Practical Example (Modern Systems)

- Wi-Fi / 4G / 5G systems dynamically switch modulation:
  - Good signal → use 256-QAM (high speed)
  - Poor signal → use QPSK/BPSK (more reliable)

19. (Difference between dB and dBm **any 5 points - 5 marks, each point -1 mark)**

dB (decibel)

Definition

dB is a relative unit used to express the ratio between two power levels.

Formula

$$dB = 10\log_{10}(P1/P2)$$

Key Points

- No absolute value → always compares two quantities
- Used for gain, loss, amplification, attenuation
- Can be positive or negative

Example

- Gain of amplifier = 20 dB
- Loss in cable = −3 dB

dBm (decibel-milliwatt)

Definition

dBm is an absolute unit used to measure power with reference to 1 milliwatt (mW).

Formula

$$dBm = 10 \log_{10}(P/1\ mW)$$

Examples

- 0 dBm = 1 mW
- 10 dBm = 10 mW
- 20 dBm = 100 mW

20. **(Comparison between** dB and dBm **any 5 points - 5 marks, each point -1 mark)**

LC Oscillator

An LC oscillator uses an inductor (L) and capacitor (C) as a tank circuit to generate oscillations.

Features

Frequency determined by LC circuit $f = 1/2\pi LC$
Examples: Hartley, Colpitts oscillators

Advantages

- Simple design
- Tunable (frequency can be varied by changing L or C)
- Suitable for RF applications

Disadvantages

- Less frequency stability
- Affected by temperature and component variations
- Lower accuracy

Crystal Oscillator

A crystal oscillator uses a quartz crystal that works on the piezoelectric effect.

Features

- Very precise natural frequency
- Crystal acts like a highly stable resonant circuit Advantages

- Excellent frequency stability
- Very high accuracy
- Low drift with temperature

Advantages

Very good frequency stability
High precision
Low power consumption

Disadvantages

- Not easily tunable
- Slightly more expensive
- Limited frequency range (fixed frequencies)